6.5 Fermentors

The main function of a fermenter is to provide a suitable environment in which an organism can efficiently produce a target product. Most of them are designed to maintain high biomass concentrations, which are essential for many fermentation processes. Fermentor design, quality of construction, mode of operation and the level of sophistication largely depend upon the production organism, the optimal operating conditions required for target product formation, product value and the scale of production. The performance of any fermenter depends on many factors, but the key physical and chemical parameters that must be controlled are agitation rate, oxygen transfer, pH, and temperature and foam production.

HOTS

What will happen if antifoam agents are not used in the Fermentation process?

6.5.1 Basic Design of a Fermenter

The materials used for construction of fermenter withstand repeated steam sterilization and are nontoxic. The reaction vessel is designed to withstand vacuum or else it may collapse while cooling. The internal surface is smooth and corrosion resistant. Either stainless steel or glass is used for construction.

Conventional bioreactors are cylindrical vessels with dome top and bottom (Figure 6.5).

It is surrounded by a jacket and sparger at the bottom through which air is introduced. The agitator (for mixing of cells and medium) shaft is connected to a motor at the bottom. (Figure 6.6) It has ports for pH, temperature, dissolved Oxygen sensors for regulation. Antifoam agents like animal vegetable oil, lard oil, corn oil and soya bean oil are used to control the foam. Modern fermentors are usually integrated with computers for efficient process monitoring and data acquisition. Parts of the fermenter and their functions are given in Table 6.2.

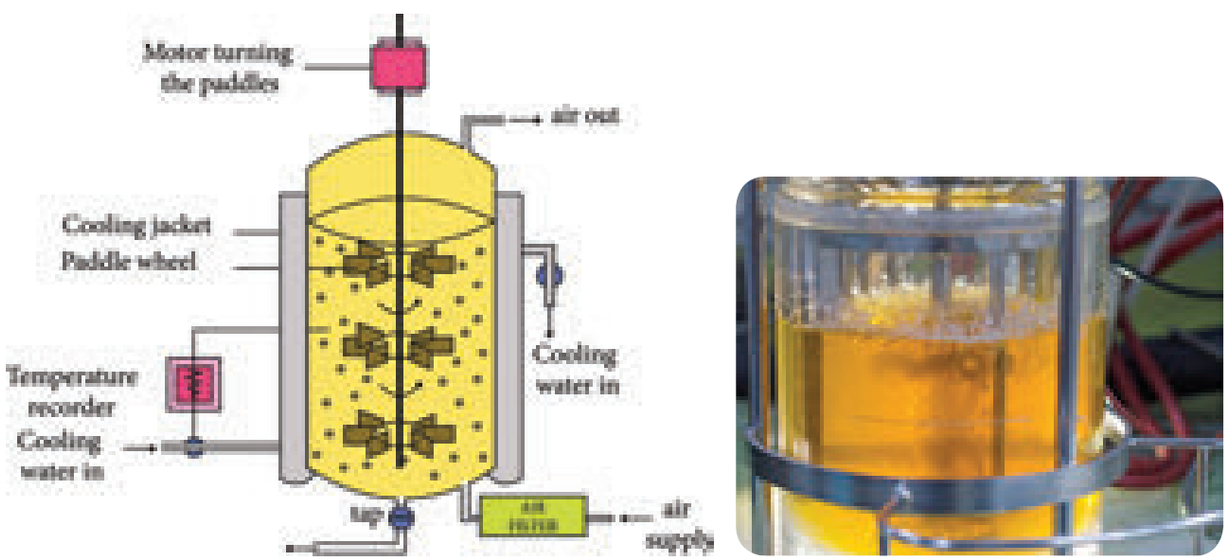


Figure 6.5: Design of a fermenter

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